Spectral Indices: Equations and Interpretations

Sentinel-2 Level-2A | Landsat 8/9 Collection 2 Level-2

16 spectral indices • Individual explanations • Sentinel-2 and Landsat 8/9 equations only

Equations for surface reflectance imagery

Each index is presented separately with a short explanation and its Sentinel-2 and Landsat 8/9 band equations. No data conversion or scaling formulas are included.

Band labels refer to the original sensor bands. Sentinel-2: B8 (NIR), B8A (narrow NIR), B11 (SWIR1), B12 (SWIR2). Landsat 8/9: B5 (NIR), B6 (SWIR1), B7 (SWIR2).

1. NDVI — Normalized Difference Vegetation Index

Measures vegetation greenness and photosynthetic activity. Higher positive values generally indicate greener vegetation.

Sentinel-2	**(B8 − B4) / (B8 + B4)**
Landsat 8/9	(B5 − B4) / (B5 + B4)

Typical application: Used to monitor vegetation cover, crop condition, and seasonal greenness.

2. GNDVI — Green Normalized Difference Vegetation Index

Uses green instead of red reflectance; can be sensitive to changes in leaf chlorophyll and canopy condition.

Sentinel-2	**(B8 − B3) / (B8 + B3)**
Landsat 8/9	(B5 − B3) / (B5 + B3)

Typical application: Useful for crop and vegetation monitoring; interpret alongside ground observations.

3. EVI — Enhanced Vegetation Index

Enhances the vegetation signal and incorporates blue reflectance to reduce some atmospheric and canopy-background effects.

Sentinel-2	**2.5(B8 − B4) / (B8 + 6B4 − 7.5B2 + 1)**
Landsat 8/9	2.5(B5 − B4) / (B5 + 6B4 − 7.5B2 + 1)

Typical application: Useful for vegetation monitoring where NDVI may saturate in dense canopies.

4. EVI2 — Two-Band Enhanced Vegetation Index

A two-band alternative to EVI that does not require the blue band.

Sentinel-2	**2.5(B8 − B4) / (B8 + 2.4B4 + 1)**
Landsat 8/9	2.5(B5 − B4) / (B5 + 2.4B4 + 1)

Typical application: Useful when a simpler vegetation index similar to EVI is desired.

5. SAVI — Soil-Adjusted Vegetation Index

Reduces the influence of exposed soil on vegetation estimates; here the soil adjustment factor L is 0.5.

Sentinel-2	**1.5(B8 − B4) / (B8 + B4 + 0.5)**
Landsat 8/9	1.5(B5 − B4) / (B5 + B4 + 0.5)

Typical application: Suitable for drylands, sparse vegetation, and agricultural fields with visible soil.

6. OSAVI — Optimized Soil-Adjusted Vegetation Index

A soil-adjusted vegetation index using a fixed correction of 0.16 to lessen soil-background effects.

Sentinel-2	**1.16(B8 − B4) / (B8 + B4 + 0.16)**
Landsat 8/9	1.16(B5 − B4) / (B5 + B4 + 0.16)

Typical application: Useful for crop monitoring and areas with partial vegetation cover.

7. NDRE — Normalized Difference Red-Edge Index

Compares narrow near-infrared and red-edge reflectance to examine canopy chlorophyll and vegetation condition.

Sentinel-2	**(B8A − B5) / (B8A + B5)**

Landsat 8/9	Not available with equivalent Landsat 8/9 OLI bands

Typical application: Useful for vegetation condition and chlorophyll-oriented analysis; Sentinel-2 uses its red-edge bands.

8. NDWI — Normalized Difference Water Index (McFeeters)

Highlights open surface water using green and near-infrared reflectance. Positive values can indicate water, but thresholds vary.

Sentinel-2	**(B3 − B8) / (B3 + B8)**
Landsat 8/9	(B3 − B5) / (B3 + B5)

Typical application: Useful for mapping rivers, lakes, and potential floodwater; validate against imagery.

9. MNDWI — Modified Normalized Difference Water Index

Uses green and SWIR1 bands to improve open-water separation, often reducing confusion with built-up surfaces.

Sentinel-2	**(B3 − B11) / (B3 + B11)**
Landsat 8/9	(B3 − B6) / (B3 + B6)

Typical application: Useful for urban water mapping and flood delineation; local threshold validation remains necessary.

10. NDMI — Normalized Difference Moisture Index

Compares near-infrared and SWIR1 reflectance to indicate vegetation/canopy moisture conditions.

Sentinel-2	**(B8 − B11) / (B8 + B11)**
Landsat 8/9	(B5 − B6) / (B5 + B6)

Typical application: Useful for vegetation moisture and drought-stress monitoring; also called Gao-type NDWI in some sources.

11. MSI — Moisture Stress Index

Expresses SWIR1 reflectance relative to near-infrared reflectance. Larger values often suggest greater vegetation moisture stress.

Sentinel-2	**B11 / B8**
Landsat 8/9	B6 / B5

Typical application: Useful for examining vegetation moisture stress; unlike normalized indices it is not restricted to −1 to +1.

12. NDBI — Normalized Difference Built-up Index

Contrasts SWIR1 and near-infrared reflectance to highlight many built-up surfaces. Bare soil can also produce high values.

Sentinel-2	**(B11 − B8) / (B11 + B8)**
Landsat 8/9	(B6 − B5) / (B6 + B5)

Typical application: Useful for exploratory urban mapping when combined with NDVI and other information.

13. BSI — Bare Soil Index

Combines SWIR1, red, NIR, and blue reflectance to emphasize exposed soil relative to vegetation and other cover.

Sentinel-2	**[(B11 + B4) − (B8 + B2)] / [(B11 + B4) + (B8 + B2)]**
Landsat 8/9	[(B6 + B4) − (B5 + B2)] / [(B6 + B4) + (B5 + B2)]

Typical application: Useful for bare-land mapping and soil-exposure studies.

14. NBR — Normalized Burn Ratio

Uses NIR and SWIR2 to distinguish unburned vegetation and burned surfaces. Differences between pre- and post-fire NBR can assess burn severity.

Sentinel-2	**(B8 − B12) / (B8 + B12)**
Landsat 8/9	(B5 − B7) / (B5 + B7)

Typical application: Useful for burned-area detection and change analysis (dNBR).

15. NBR2 — Normalized Burn Ratio 2

Uses SWIR1 and SWIR2; sensitive to variations in moisture and burned or exposed surfaces.

Sentinel-2	**(B11 − B12) / (B11 + B12)**
Landsat 8/9	(B6 − B7) / (B6 + B7)

Typical application: Useful as a complementary index for burned-area and moisture analysis.

16. NDSI — Normalized Difference Snow Index

Contrasts green and SWIR1 reflectance to identify snow and ice. Its algebraic form matches MNDWI, but application and classification differ.

Sentinel-2	**(B3 − B11) / (B3 + B11)**
Landsat 8/9	(B3 − B6) / (B3 + B6)

Typical application: Useful for snow mapping where snow is present; not generally relevant to lowland Somalia.

Interpretation notes

- Normalized-difference indices generally range from −1 to +1 for nonnegative reflectances with valid denominators. MSI is not restricted to this range.
- A spectral index is not itself a definitive land-cover classification. Thresholds depend on sensor, surface type, season, atmospheric conditions, and local validation.
- For flood mapping, compare cloud-masked, co-registered before-flood and during-flood water masks. An annual median composite can conceal short-lived floodwater.